

YASHVINI VARMA

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EDUCATION

B.Tech in Electronics and Communication Engineering

Sri Aurobindo Institute of Technology, Indore

Expected Graduation: 2028 | Current CGPA: 8.38

TECHNICAL SKILLS

- **Hardware Description Languages & EDA:** TL-Verilog, Verilog, Makerchip IDE, KiCad, Proteus
- **Programming & Environments:** Git/Github, C/C++, Python, Linux/Bash
- **Domains:** Digital Logic Design, RTL Design & Simulation, PCB Schematic & Layout, Signal Processing Fundamentals

EXPERIENCE & OPEN-SOURCE CONTRIBUTIONS

Redwood EDA / Makerchip Community | Open-Source Curriculum Developer

May 2026 - Present

- Collaborated directly with Redwood EDA leadership to develop and refine structured educational modules for TL-Verilog and digital architecture curriculums. ([Repo Link: Github](#))
- Authored clean, synthesizable TL-Verilog examples and visual simulation modules (VIZ) to teach pipeline logic, instruction decoding, and sequential circuit design.

KEY HARDWARE & RTL PROJECTS

RISC-V Instruction Decoder & Microarchitecture Blocks | TL-Verilog, Makerchip IDE

- Designed and simulated an instruction decode stage for the RISC-V ISA targeting opcode, register, and immediate extraction.
- Built Makerchip visualization components for cycle-level observation of datapath/control behavior.
- Implemented core combinational and sequential building blocks including parameterized nested multiplexers and a 4-bit Linear Feedback Shift Register (LFSR).
- Github Repo: <https://github.com/Spy3110/Makerchip-VIZ-Prototypes>

SilicOn_Heartbeat (555 Timer PCB) | KiCad

- Engineered a complete hardware PCB layout based on the NE555 timer IC configured in astable multivibrator mode.
- Executed end-to-end electronic design workflow: schematic capture, component footprint assignment, strict DRC (Design Rule Checking), and manufacturing Gerber generation.
- Github Repo: https://github.com/Spy3110/SilicOn_Heartbeat-555-PCB

Bridge Rectifier & Power Regulation Board | KiCad, Circuit Simulation

- Designed an AC-to-DC conversion circuit featuring full-wave bridge rectification, capacitive ripple filtering, and output regulation.
- Conducted PCB trace width optimization to handle power delivery and verified component tolerances for thermal stability.
- Github Repo: <https://github.com/Spy3110/Orange-Rectifier-Kit-PCB>

CERTIFICATIONS & MILESTONES

- NPTEL SWAYAM Silver Elite: Signals and Systems | 78% | April 2026
- Aspire Leaders Program: Global Leadership Development (Cohort 5, 2025)